Technical & Functional Specification Document

Bluetooth Test Infrastructure Dashboard Agent (Claude Code Edition)

1. Overview & Purpose

This document specifies the requirements, architecture, test plan structure, data models, and LLM-driven core logic for an AI-powered **Bluetooth Test Infrastructure Dashboard Agent**. The primary objective of the Dashboard Agent is to consume multi-source test execution logs and structured telemetry from automated Bluetooth silicon/platform validation runs, synthesize complex test results into actionable executive insights, detect anomalies, conduct semantic bug clustering, calculate milestone/release risks, and automatically generate prioritized next actions. This project will not have any real systems and test infrastructure. Instead a random data set will need to be generated and the Agent will analyse the data set and share the dashboard.

2. Test Execution Environment & Test Plan Specification

The testing pipeline validates a custom **Bluetooth Chipset for Mobile Platforms** across four primary test domains.

2.1 Test Domains & Baseline Metrics

Test Domain	Scope & Focus	Total Test Cases / Setup	Full Cycle Execution Duration	Failure / Crash Behavior	Jira Automation
1. HCI Regression	Layer-level HCI commands, state transitions, protocol compliance up to Bluetooth v5.2.	5,000 test cases	7 Days	Failures extend duration; automatic retry/re-execution log logging.	100% Automated ticketing & mapping.

2. Phone Testing	Interoperability & high-level BT profiles/features on mobile platform UI/kernel.	1,000 test cases	1 Day	Retry on transience; direct bug tracking.	100% Automated ticketing & mapping.
3. HCI MTBF	Stress & reliability testing across platform stability runs.	16 Device Under Test (DUT) nodes running for 5×24 hours (120 continuous hrs).	5 Days	Catches system crashes/hangs only. Auto-recovery and restart post-crash with timestamp logging.	Auto-de-duplication & stack trace mapping to unique Jiras.
4. KPI & Performance	Audio streaming quality (A2DP, LE Audio), latency, throughput, connection setup time, power consumption.	Feature-based KPI matrix	3 Days	Threshold breach alerts; degraded trend analysis.	Auto-generated performance bug reports.

2.2 Execution Constraints & Schedule

- **Full Cycle Duration:** Standard run takes **7 calendar days** when run in parallel across the

test bed infrastructure.

- **Elastic Duration Rule:** High failure rates increase total cycle time due to automated re-runs, diagnostic log collection, and dynamic isolation suites.
- **Environment Stability Assumption:** Test bench setup/hardware flakiness is handled upstream; test failures reflect true chip/stack defect behavior.

3. Module 1: Sample Test Run Data Generator (Mock Data Engine)

To power the Claude Code implementation and unit testing, a deterministic mock data generator module (SampleTestRunGenerator) produces comprehensive execution telemetry in JSON format. To Make this More real, There needs to be a one time test generator , which will actually generate this test plans completely. So that, the random test run data generated will have details of what test cases also got failed in each test suite.

3.1 Data Schema Specification(Below is the Sample, please add more entries that are required in order to have a Smart and intelligent Dashboard which provides best output.

```
JSON
{
 "test_run_metadata": {
  "run_id": "BT_FULL_CYCLE_20261020_01",
  "chipset_revision": "BT_5_2_REV_C2",
  "firmware_build": "FW_v14.2.109_RC3",
  "start_time": "2026-10-13T00:00:00Z",
  "end_time": "2026-10-20T11:56:00Z",
  "cycle_type": "FULL_CYCLE"
 },
 "domain_summaries": [
  {
   "domain": "HCI_REGRESSION",
   "total_tests": 5000,
   "passed": 4820,
   "failed": 120,
   "blocked": 60,
   "execution_duration_hours": 168,
   "feature_breakdown": {
    "LE_Advertising": {"pass": xx, "fail": 10, "blocked": 0},
    "LE_Audio_CIS_BIS": {"pass": xx, "fail": 75, "blocked": 45},
    "BR_EDR_Connections": {"pass": xx, "fail": 15, "blocked": 5},
    "GATT_ATT": {"pass": xx, "fail": 20, "blocked": 10}
```

Like this , there needs to be a total of 25 features.


```
    }
   },
   {
    "domain": "PHONE_TESTING",
    "total_tests": 1000,
    "passed": 940,
    "failed": 45,
    "blocked": 15,
    "execution_duration_hours": 24
   "feature_breakdown": {
     "LE_Advertising": {"pass": xx, "fail": 10, "blocked": 0},
     "LE_Audio_CIS_BIS": {"pass": xx, "fail": 75, "blocked": 45},
     "BR_EDR_Connections": {"pass": xx, "fail": 15, "blocked": 5},
     "GATT_ATT": {"pass": xx, "fail": 20, "blocked": 10}
```


Like this , there needs to be a total of 15 features.


```
   },
   {
    "domain": "MTBF",
    "duts_active": 16,
    "planned_hours_per_dut": 120,
    "total_operational_hours": 1920,
    "total_crashes": 14,
    "unique_crashes": 3,
    "recovery_success_rate_percent": 100
   },
   {
    "domain": "KPI_AUDIO",
    "metrics": [
     {"metric_name": "A2DP_Latency_ms", "target": "< 100", "measured_avg": "125", "status":
"DEGRADED"},
     {"metric_name": "LE_Audio_THD", "target": "< 0.5%", "measured_avg": "0.3%", "status": "PASS"},
     {"metric_name": "Connect_Time_ms", "target": "< 400", "measured_avg": "380", "status": "PASS"}
    ]
   }
 ],
 "jira_telemetry": {
  "total_tickets_linked": 179,
  "severity_distribution": {
   "Blocker": 4,
   "Critical": 28,
```

```
    "Medium": 102,
    "Trivial": 45
   },
   "crash_mapping": [ (Below is a sample with only 3 crashes, But in general, there will be at least 10 crashes in the initial stage which might occur multiple times, and later period it can come down to zero.
    {
     "jira_id": "BTFW-4012",
     "crash_signature": "NullPointerInLC3CodecBuffer",
     "occurrences": 8,
     "affected_duts": ["DUT-02", "DUT-05", "DUT-09", "DUT-14"],
     "severity": "Blocker"
    },
    {
     "jira_id": "BTFW-4018",
     "crash_signature": "LinkManagerTimeoutOnRoleSwitch",
     "occurrences": 4,
     "affected_duts": ["DUT-01", "DUT-07"],
     "severity": "Critical"
    },
    {
     "jira_id": "BTFW-4022",
     "crash_signature": "HCI_Reset_Hang_Post_A2DP_Stream",
     "occurrences": 2,
     "affected_duts": ["DUT-11"],
     "severity": "Critical"
    }
   ]
  }
}
```


4.1 Specification of the 5 Core Intelligence Outputs

1. Automated Insight Synthesis ("TL;DR")

- **Goal:** High-level summary for Engineering Managers & Release Leads.
- **Requirements:**
  - Summarize pass rate across all all test cases.
  - State total cycle duration (and delta from baseline 7 days due to failure re-runs).
  - High-level health index score (0–100%) for the current build.

2. Risk Assessment & Milestone Forecasting

- **Goal:** Evaluate build readiness against target tape-out or release dates.

- **Requirements:**
  - Evaluate **Blocker** & **Critical** Jiras against release gates.
  - Compute schedule slippage estimates based on failure volumes.
  - Evaluate MTBF stability (e.g., Mean Time Between Failures in hours per DUT).

3. Anomaly Detection & Blindspot Flagging

- **Goal:** Uncover non-obvious defects, hardware-dependent issues, and untested gaps.
- **Requirements:**
  - Flag DUT specific variance (e.g., *DUT-02 failed 3x more crashes than DUT-03*).
  - Identify feature degradation cross-talk (e.g., LE Audio failure spikes matching A2DP latency degradation).
  - Highlight "Blocked" test case concentrations indicative of gating driver bugs.

4. Semantic Bug Clustering (Actionable Grouping)

- **Goal:** Group hundreds of raw failures/Jiras into root-cause buckets for dev teams.
- **Requirements:**
  - Cluster Jiras by underlying firmware subsystem (e.g., *Host-Controller Interface*, *Audio Codec / LC3*, *Link Manager Protocol*, *PHY Layer*).
  - Highlight recurring crash signatures vs single-instance failures.

5. Automated "Next Actions" Generation

- **Goal:** Provide prescriptive, task-oriented recommendations assigned by developer role.
- **Requirements:**
  - Output targeted tasks (e.g., *Firmware Team:* Fix BTFW-4012 LC3 buffer overflow immediately to unblock 45 blocked LE Audio tests).
  - Recommend specific re-run subsets rather than full cycle executions.